COURIER MANAGEMENT SYSTEM(CMS)

User Story:

Courier and logistics companies often struggle to efficiently manage shipments and deliveries due to reliance on manual processes and disconnected systems. The absence of a centralized shipment tracking mechanism makes it difficult to monitor parcel movement in real time, resulting in delayed delivery updates and poor visibility for customers. Delivery agent assignments are performed manually, increasing the chances of scheduling conflicts and operational delays. Furthermore, the lack of automated notifications prevents timely communication with customers, while the absence of performance analytics limits managerial insight into operational efficiency. Managing multiple branches becomes complex without an integrated platform, leading to coordination issues across locations. Collectively, these challenges result in customer dissatisfaction, reduced productivity, operational inefficiencies, and ultimately revenue loss for the organization.

1.INTRODUCTION

1.1 Project Overview :

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies. The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

1.2 Purpose :

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

2.2 Empathy Map Canvas

User Personal: Branch Manager

- **Think & Feel:** Concerned about delays & errors
- **Hear:** Customer complaints
- **See:** Registers & spreadsheets
- **Say & Do:** Manual follow-ups
- **Pain:** Delivery conflicts & missed updates
- **Gain:** Automated tracking & dashboards

2.3 Brainstorming & Idea Prioritization

Ideas Considered

● Manual registers

● Spreadsheet tracking

● Standalone portal

● Salesforce CRM

Final Selection

Salesforce CRM selected due to:

● Automation

● Scalability

● Real-time visibility

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

● Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting

Functional Requirements

1. Create and manage shipment records
2. Store source and destination address
3. Assign delivery agents to shipments
4. Update shipment status
5. Track delivery progress

Non-Functional Requirements

1. Easy user interface
2. Secure data storage
3. Fast processing
4. Reliable system performance

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

4.PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce

4.2Solution Architecture

Components:
● Customer Object
● Shipment Object
● Delivery Agent Object
● Branch Object
● Delivery Update Object
● Invoice Object
● Flows
● Reports
● Dashboardsplatform.

5.PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

This phase defines how the project will be developed step-by-step.

Project Milestones

1. Salesforce Account Setup
2. Object Creation
3. Tabs Creation
4. Fields & Relationships
5. Validation Rules
6.Page Layouts

6.Project Development Phase:

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place.

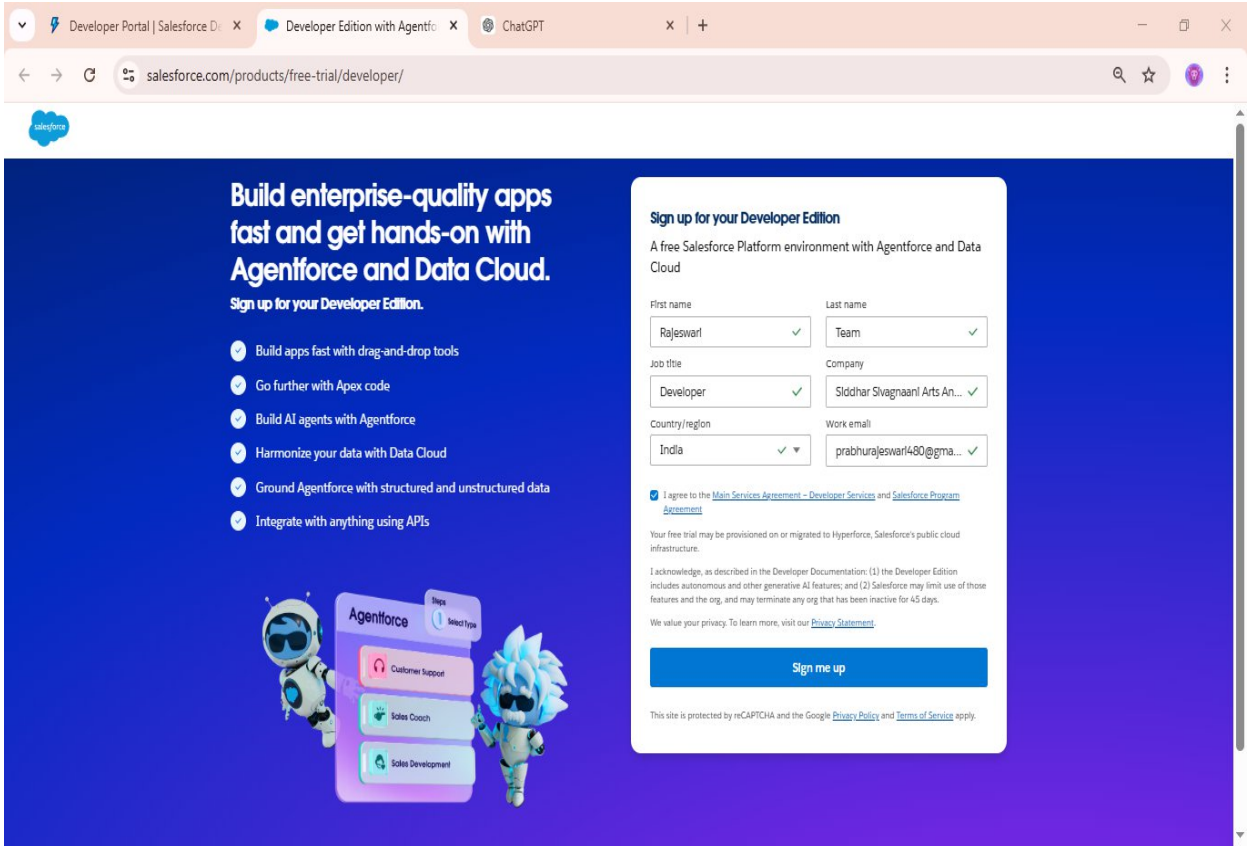
What Is Salesforce ?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

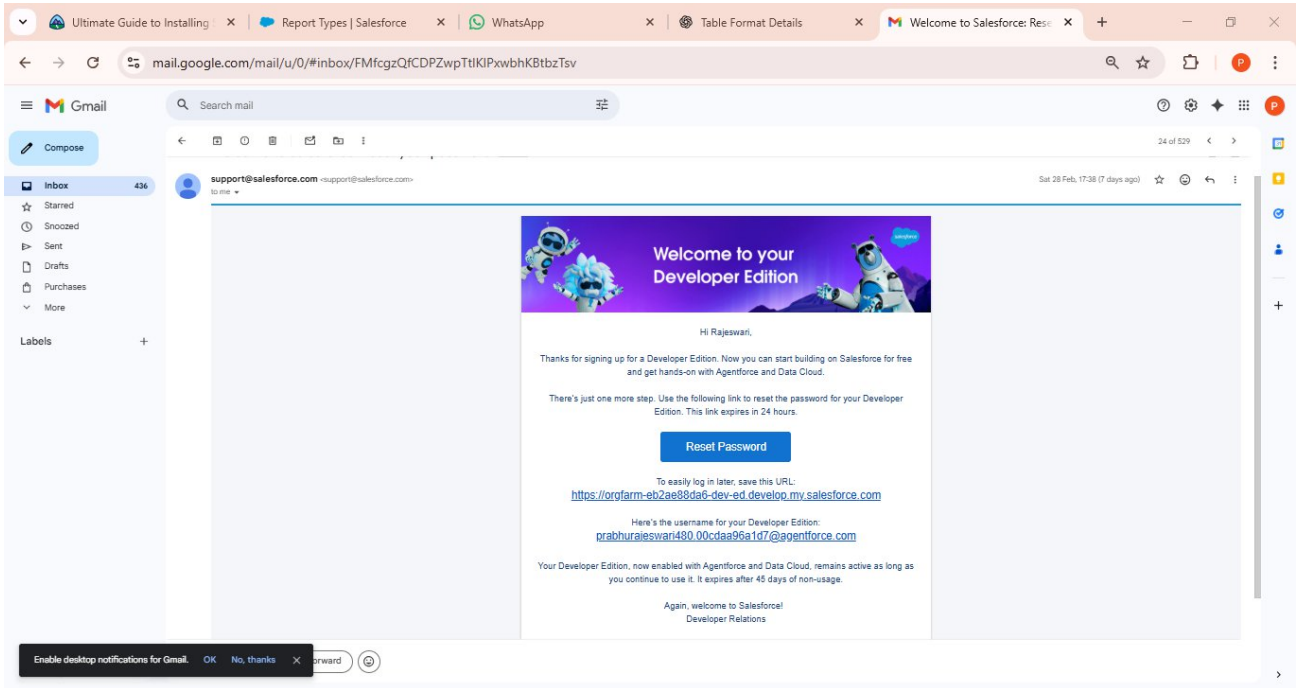
Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email



Activity 2: Account Activation



Milestone 2- Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information

2. **Shipment__c** – manages parcel details, source, destination, and tracking information

3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details

4.**Branch__c** – represents courier branch offices.

5. **Delivery_Update__c** – stores status updates for each shipment

6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

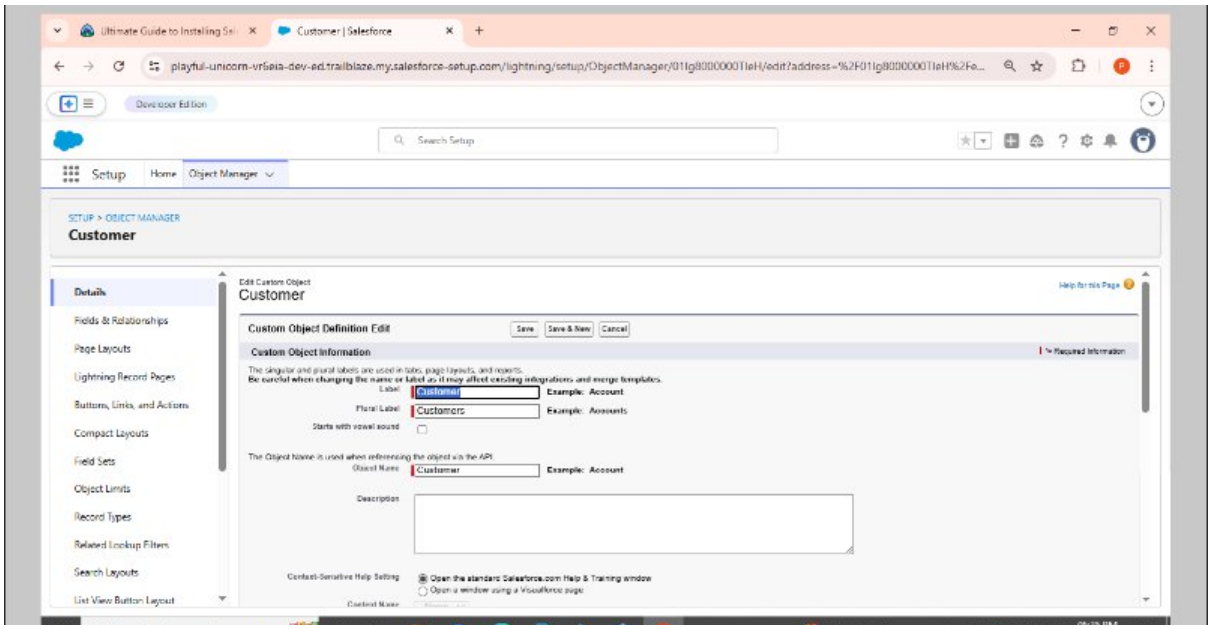
Delivery_Agent → Branch (Lookup)

Delivery_Update → Shipment (Master-Detail)

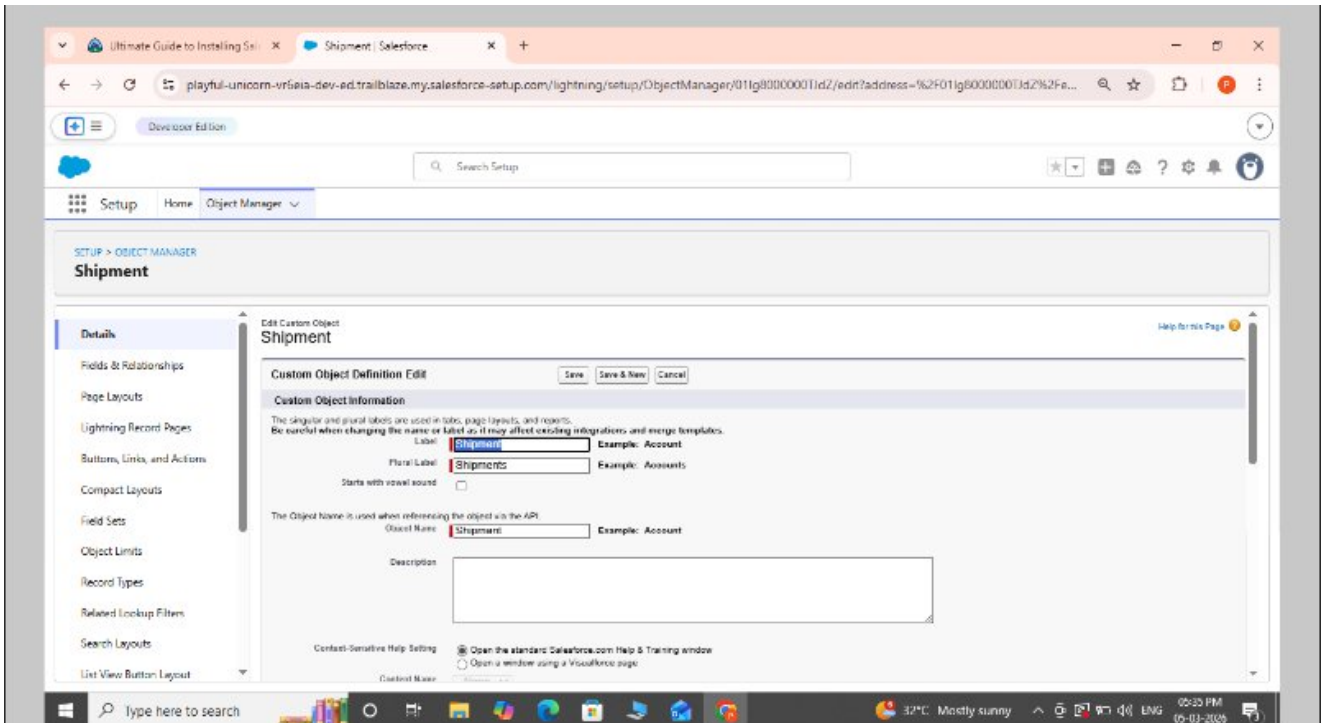
Invoice → Shipment (Lookup)

Steps to Implement :

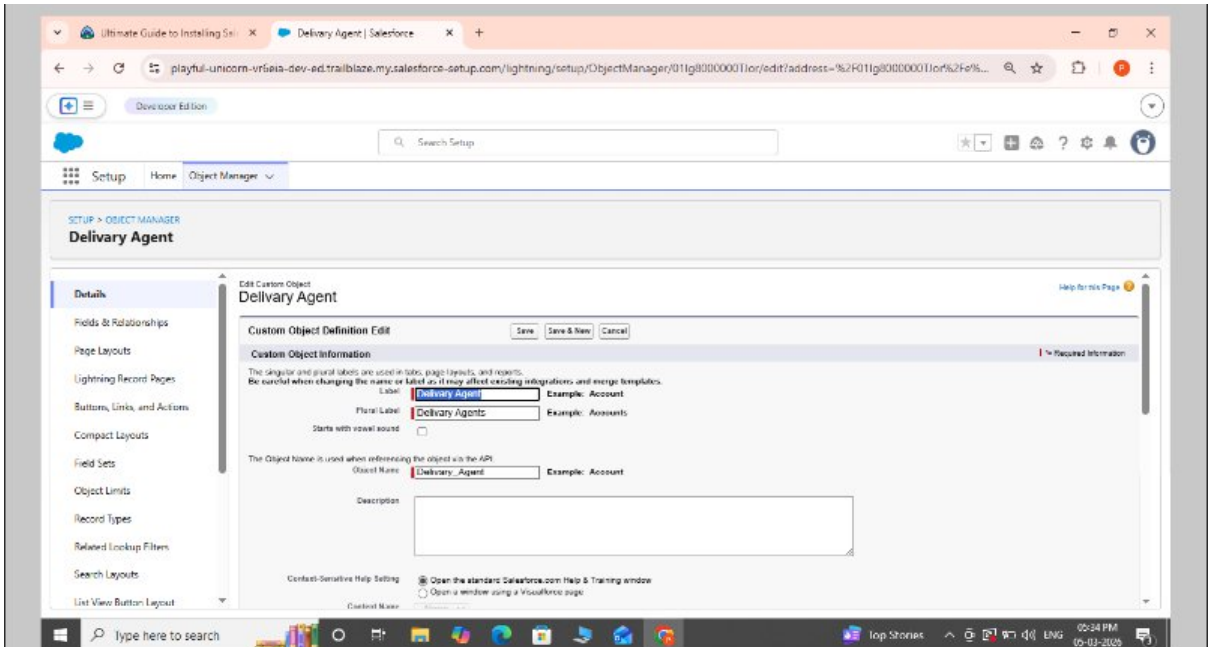
Activity 1: Create Customer Object



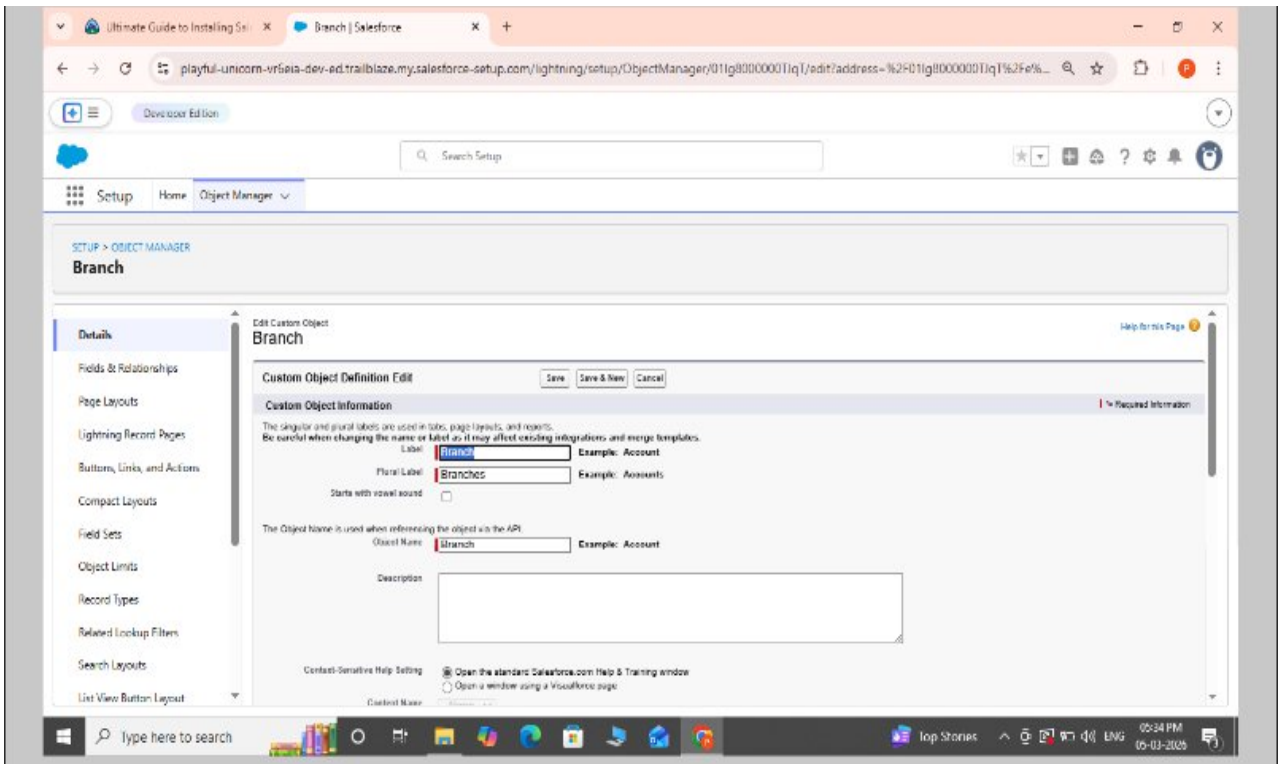
Activity 2: Create Shipment Object



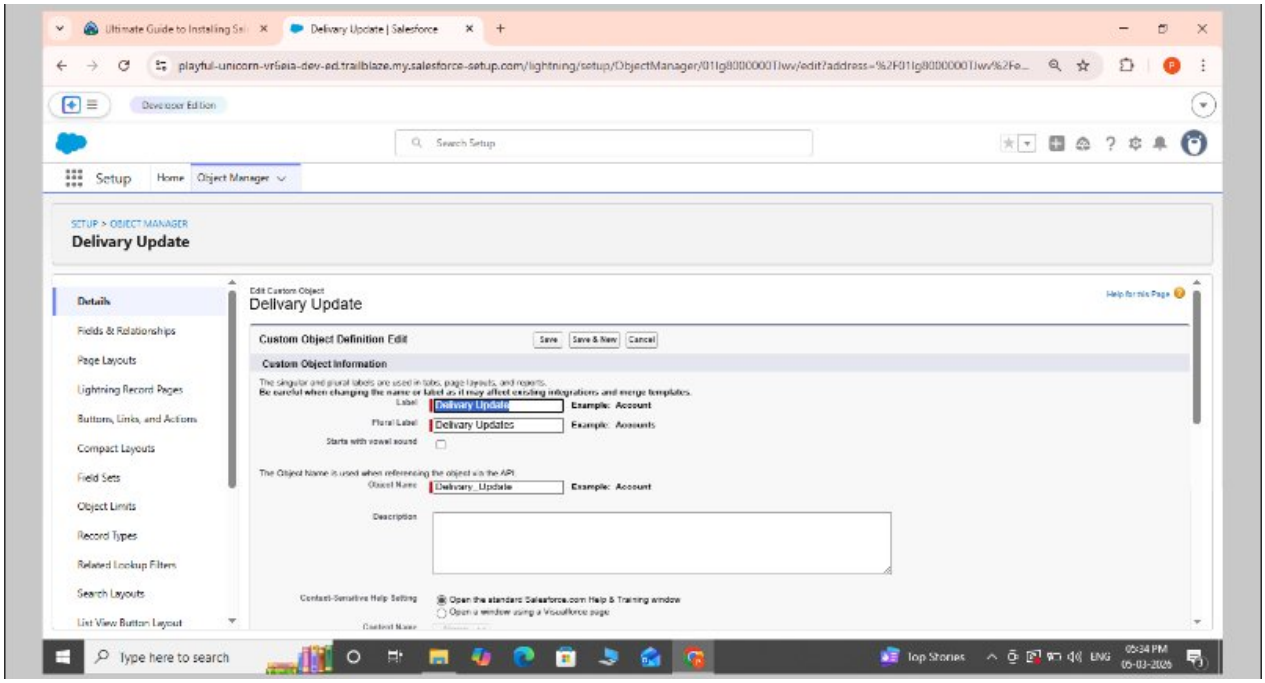
Activity 3: Create Delivery_Agent Object



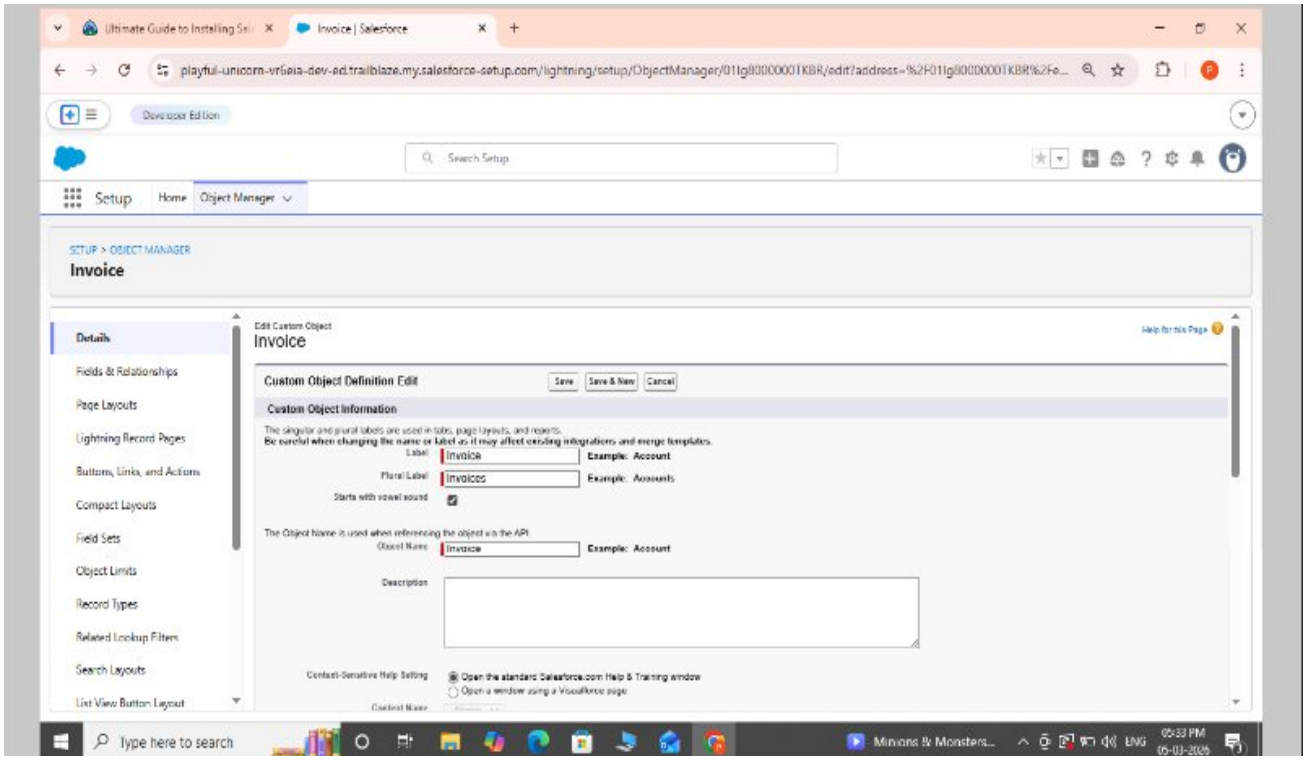
Activity 4: Create Branch Object



Activity 5: Create Delivery_Update Object



Activity 6: Create Invoice Object



Milestone 3 - Tabs

Tab:

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

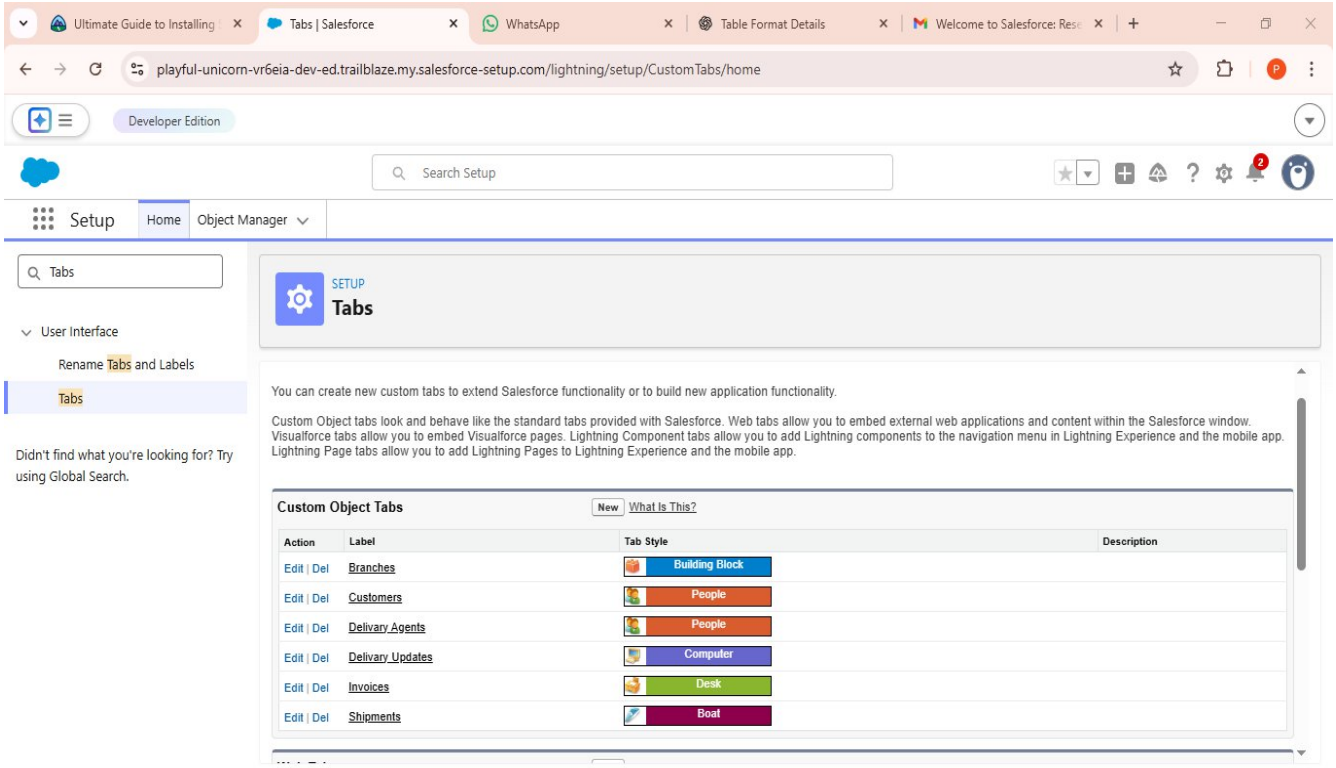
Activity 1: Creating a Tab for Customer Object

Activity 2: Creating a Tab for Shipment Object

Activity 3: Creating a Tab for Delivery Agent Object

Activity 4: Creating a Tab for Branch Object

Activity 6: Creating a Tab for Invoice Object



Milestone 4: Fields & Relationships

What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record

Activity 1: Creation of fields for the Customer Object

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of the customer
Email	Email__c	Email	Yes	Customer email address
Phone	Phone__c	Phone	Yes	Customer contact number
Address	Address__c	Text Area	Yes	Customer address

Field Name	API Name	Data Type	Required	Description
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state
Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

1. Email (Email Field)

● Go to Setup → Object Manager → Customer__c

● Click Fields & Relationships → New

● Select Email → Next

● Field Label: Email

● Check the Required box

● Click Next → Next → Save

2. Phone (Phone Field)

● Setup → Object Manager → Customer__c

● Fields & Relationships → New

● Select Phone → Next

● Field Label: Phone

● Check Required

● Click Next → Next → Save

3. Address (Text Area)

● Setup → Object Manager → Customer__c● Fields & Relationships → New

● Select Textarea → Next

● Field Label: Address

● Check Required

● Save

4. Customer Type (Picklist)

● Setup → Customer__c → Fields → New

● Picklist → Next

● Field Label: Customer Type

● Enter values:

○ Regular

○ Business

● Check Required

● Save

5. City (Text)

● Setup → Customer__c → Fields → New

● Text → Next

● Field Label: **City**

● Save

6. State (Text)● Setup → Customer__c → Fields → New

● Text → Next

● Field Label: **State**

● Save

7. Pincode (Text)

● Setup → Customer__c → Fields → New

● Text → Next

● Field Label: **Pincode**

● Save

8. Is Active (Checkbox)

● Setup → Customer__c → Fields → New

● Checkbox → Next

● Field Label: **Is Active**

● Save

Activity 2: Creation of fields for the Shipment Object

Field Name	API Name	Data Type	Required	Description
Tracking Number	Name	Auto Number	Yes	Unique shipment ID
Customer	Customer__c	Lookup (Customer)	Yes	Related customer
Pickup Address	Pickup_Address__c	Text Area	Yes	Parcel pickup location
Delivery Address	Delivery_Address__c	Text Area	Yes	Parcel delivery location
Parcel Weight	Parcel_Weight__c	Number	Yes	Weight of the parcel
Delivery Charge	Delivery_Charge__c	Currency	Yes	Shipping cost
Shipment Status	Shipment_Status__c	Picklist	Yes	Current shipment status
Expected Delivery Date	Expected_Delivery_Date__c	Date	Yes	Estimated delivery date
Delivery Agent	Delivery_Agent__c	Lookup (Agent)	No	Assigned delivery agent
Shipment Category	Shipment_Category__c	Picklist	Yes	Domestic / International

Activity 3: Creation of fields for Delivery_Agent Object

1. Email

● Setup → Delivery_Agent__c → Fields → New

● Email → Next

● Field Label: Email

● Required → Save

2. Phone

● Setup → Delivery_Agent__c → Fields → New

● Phone → Next

● Field Label: Phone

● Required → Save

3. Status (Picklist)

Values:

● Active

● Inactive

● On Leave

Required → Save

5.Joining Date

- Setup → Delivery_Agent__c → Fields → New

● Date → Next

● Field Label: Joining Date

● Save

Activity 4: Creation of fields for Branch Object

Field Name	API Name	Data Type	Required
Office Branch Name	Name	Text	Yes
Branch ID	Branch_ID__c	Text	Yes
Location City	Location_City__c	Text	Yes
Region	Region__c	Text	Yes
Supervisor Name	Supervisor_Name__c	Text	No

Activity 5: Creation of fields for Delivery_Update Object

Field Name	API Name	Data Type	Required
Update ID	Name	Auto Number	Yes
Courier Shipment	Courier_Shipment__c	Master-Detail	Yes
Delivery Status	Delivery_Status__c	Picklist	Yes
Status Update Time	Status_Update_Time__c	Date/Time	Yes
Comments	Comments__c	Text	No

Activity 6: Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Bill Number	Name	Auto Number	Yes
Shipment Reference	Shipment_Ref__c	Lookup	Yes
Total Cost	Total_Cost__c	Currency	Yes
Billing Date	Billing_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Relationships :

Relationship Creation Steps :

Activity 1 – Shipment → Customer

● Setup → Shipment__c → Fields → New

● Lookup Relationship

● Related Object: **Customer__c**

● Field Label: **Customer**

● Save

Activity 2 – Shipment → Delivery Agent

● Setup → Shipment__c

● New → Lookup

● Object: **Delivery_Agent__c**

● Label: **Assigned Agent**

● Save

Activity 3 – Delivery Agent → Branch

● Setup → Delivery_Agent__c

● New → Lookup

● Related Object: **Branch__c**

● Label: **Branch**

● Save

Activity 4 – Delivery Update → Shipment

● Setup → Delivery_Update__c

● New → **Master-Detail**

● Related Object: **Shipment__c**

● Save

Activity 5 – Invoice → Shipment

● Setup → Invoice__c● New → Lookup

● Object: **Shipment__c**

● Save

Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Activity 1: Create Validation Rules on Customer__c

1. Validation Rule: Email Must Not Be Blank
2. Validation Rule: Phone Must Be 10 Digits
3. Validation Rule: Phone Must Be 10 Digits

Activity 2: Create Validation Rules on Shipment__c

1. Validation Rule: Weight Must Be Greater Than Zero
2. Validation Rule: Price Must Be Greater Than Zero
3. Validation Rule: Expected Date Cannot Be in the Past

Activity 3: Create Validation Rules on Delivery_Agent__c

1. Validation Rule: Email Must Be Valid

Activity 4: Create Validation Rules on Branch__c

1. Validation Rule: Branch Code Must Not Be Blank

Activity 5: Create Validation Rules on Delivery_Update__c

1. Validation Rule: Remarks Required If Status is Delayed

Milestone 6 - Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized. They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way. By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

1: Edit Page Layout for Customer__c

2: Edit Page Layout for Shipment__c

3: Edit Page Layout for Delivery_Agent__c

4: Edit Page Layout for Branch__c

5: Edit Page Layout for Delivery_Update__c

6: Edit Page Layout for Invoice__c

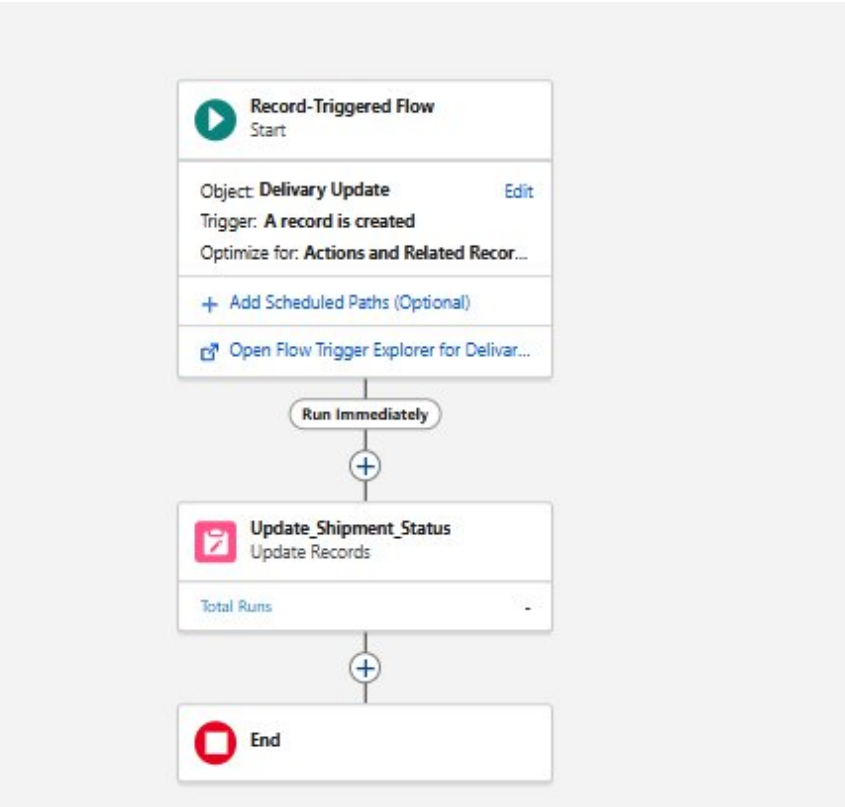
Milestone 7 - Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

Scenario:

Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update



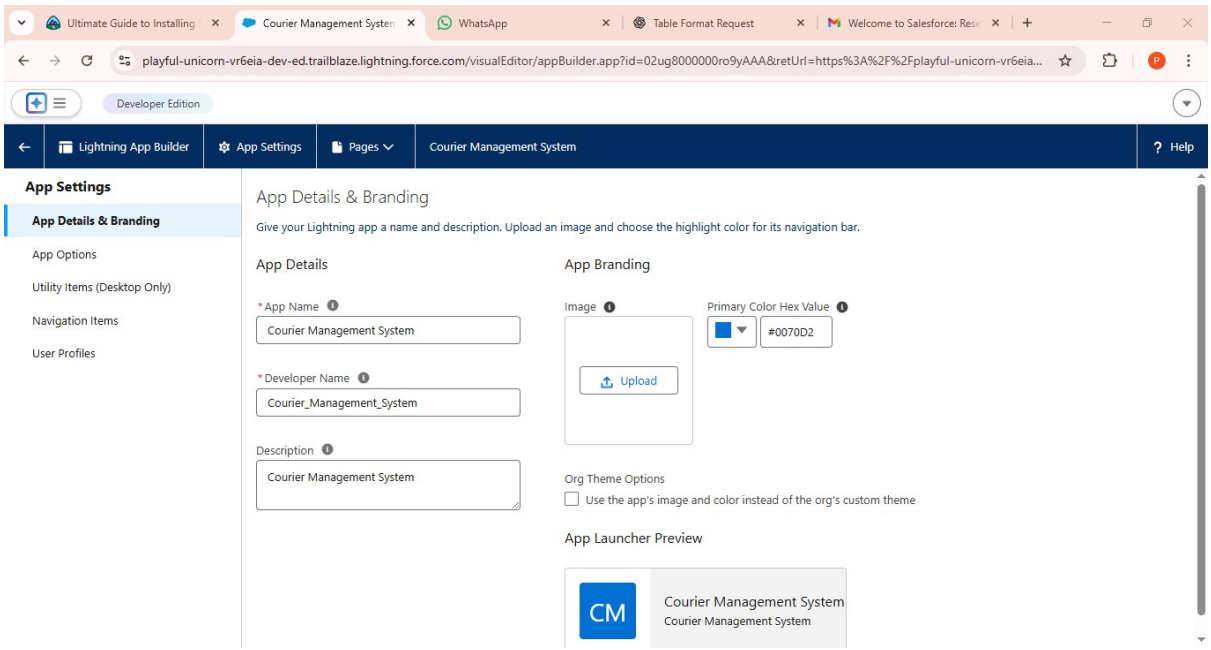
Milestone 8 - The Lightning App :

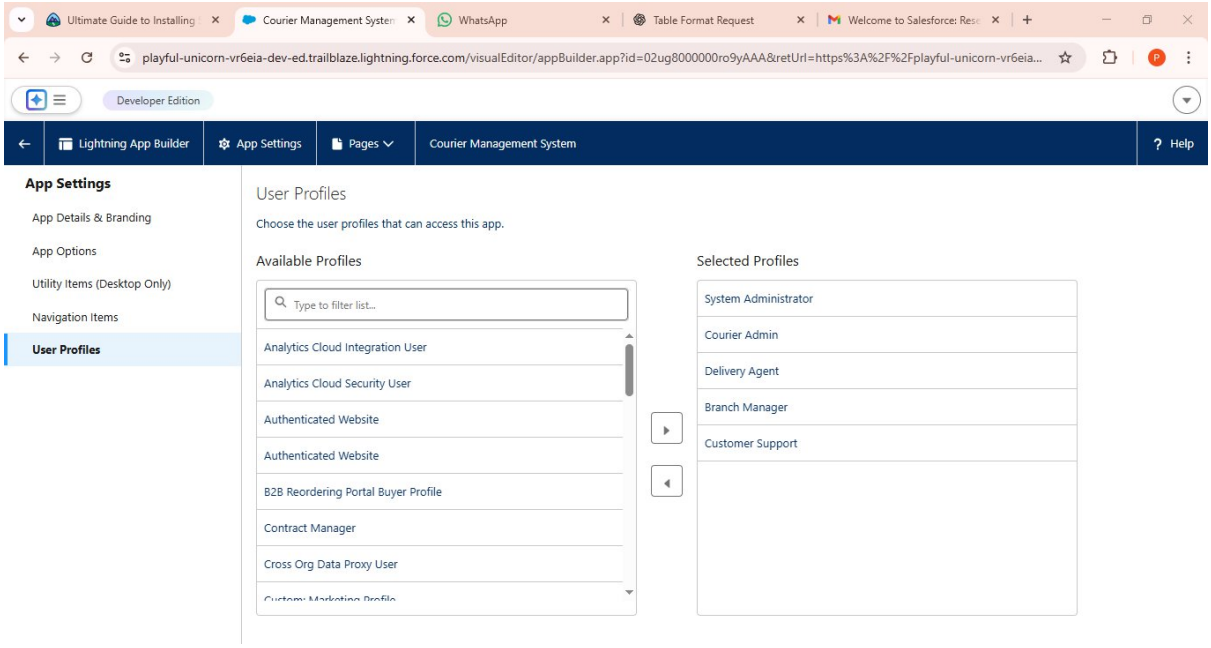
What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Create a Lightning App

Create the Courier Management System App



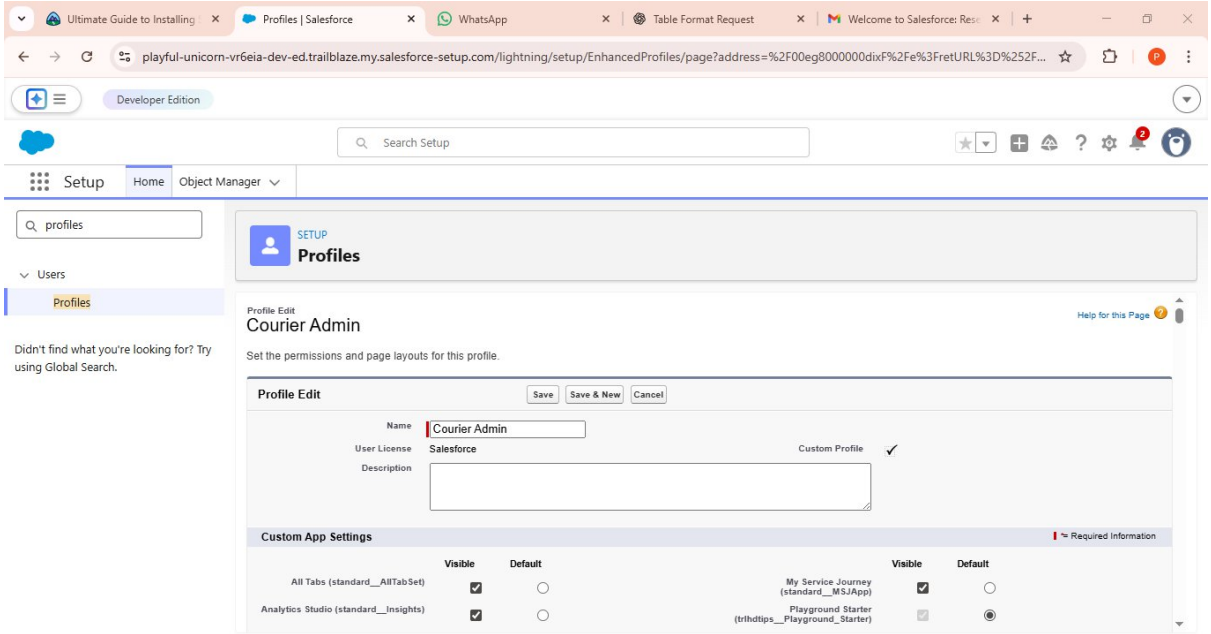


Milestone 9- Profile:

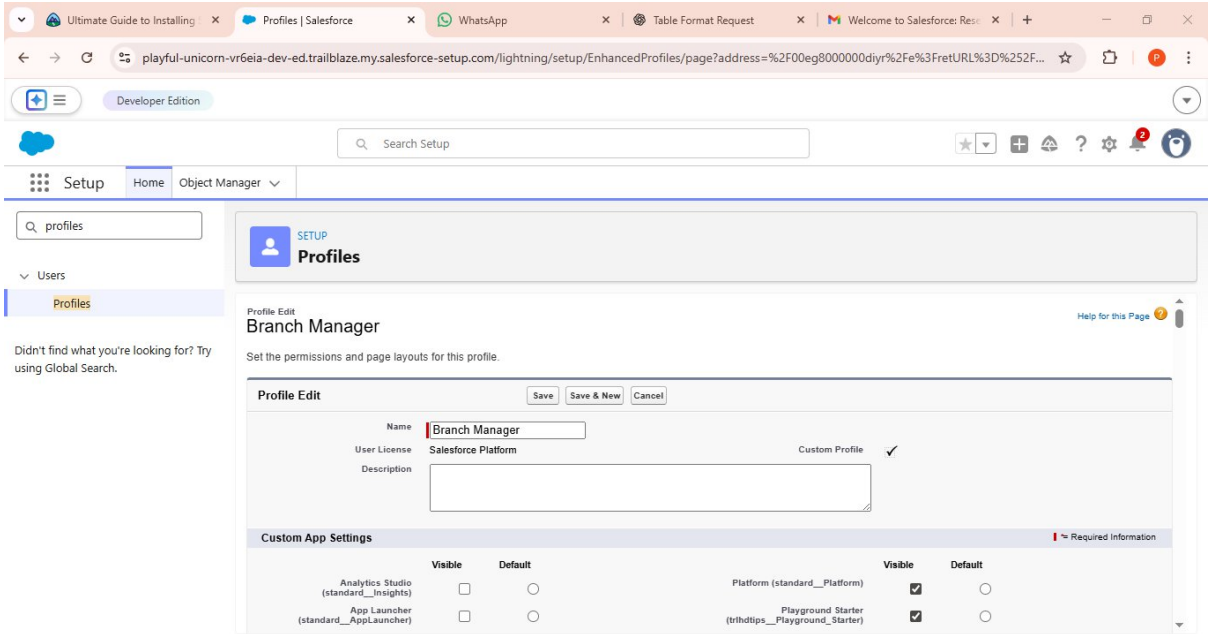
What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.A profile can be assigned to many users, but users can be assigned a single profile at a time.

Activity 1: Create a Custom Profile – Courier Admin



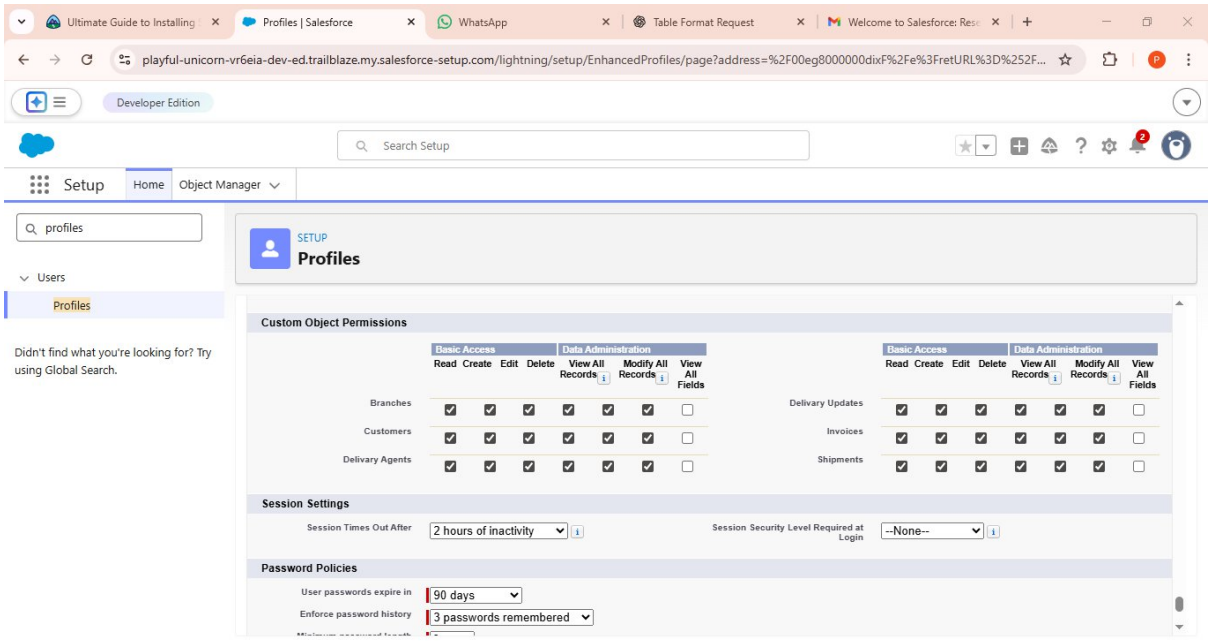
Activity 2: Create a Custom Profile – Branch Manager



Object	Access Level
Customer__c	Create, Read, Update
Shipment__c	Create, Read, Update
Delivery_Agent__c	Read, Update
Branch__c	Read, Update
Delivery_Update__c	Create, Read, Update
Invoice__c	Read
Payment__c	Create, Read
Tracking__c	Read, Update

Activity 3: Create a Custom Profile – Delivery Agent

Activity 4: Create a Custom Profile – Customer Support



Profile Name	Purpose
Courier Admin	Full system control
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking

Milestone 10- Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of Courier Admin Role

Activity 2: Creating Branch Manager Role Under Courier Admin Role

Activity 3: Creating Delivery Agent Role Under Branch Manager Role

Activity 4: Creating Customer Support Role Under Branch Manager

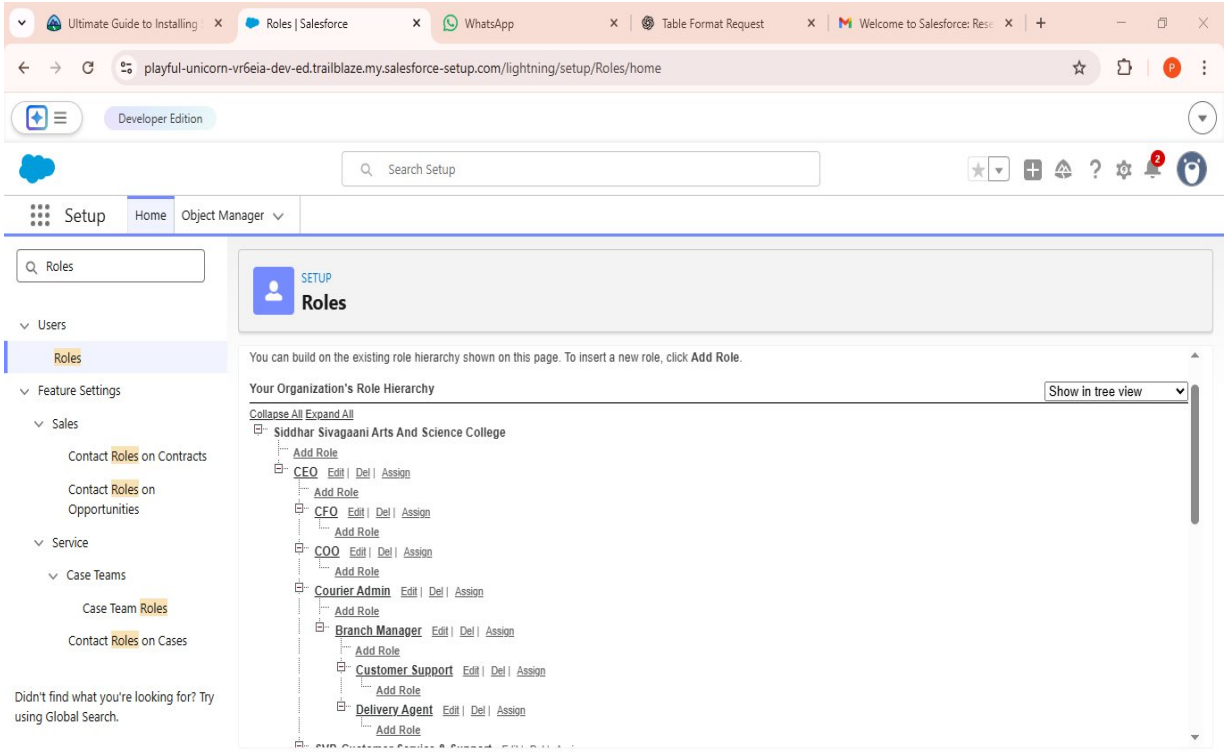
Purpose of This Hierarchy :

● **Courier Admin** – Highest level access to all data

● **Branch Manager** – Manages branch-level operations

● **Delivery Agent** – Access only to assigned shipments

● **Customer Support** – Limited access to customer and shipment tracking

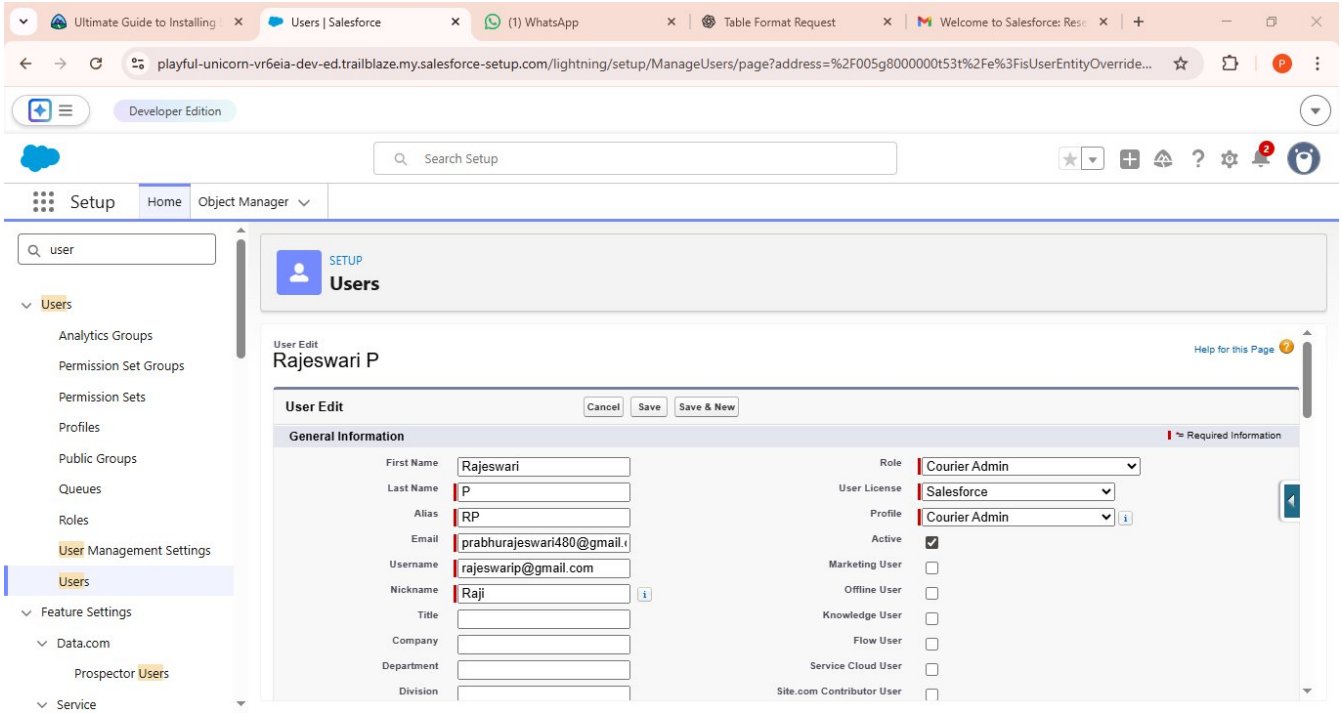


Milestone 11- User :

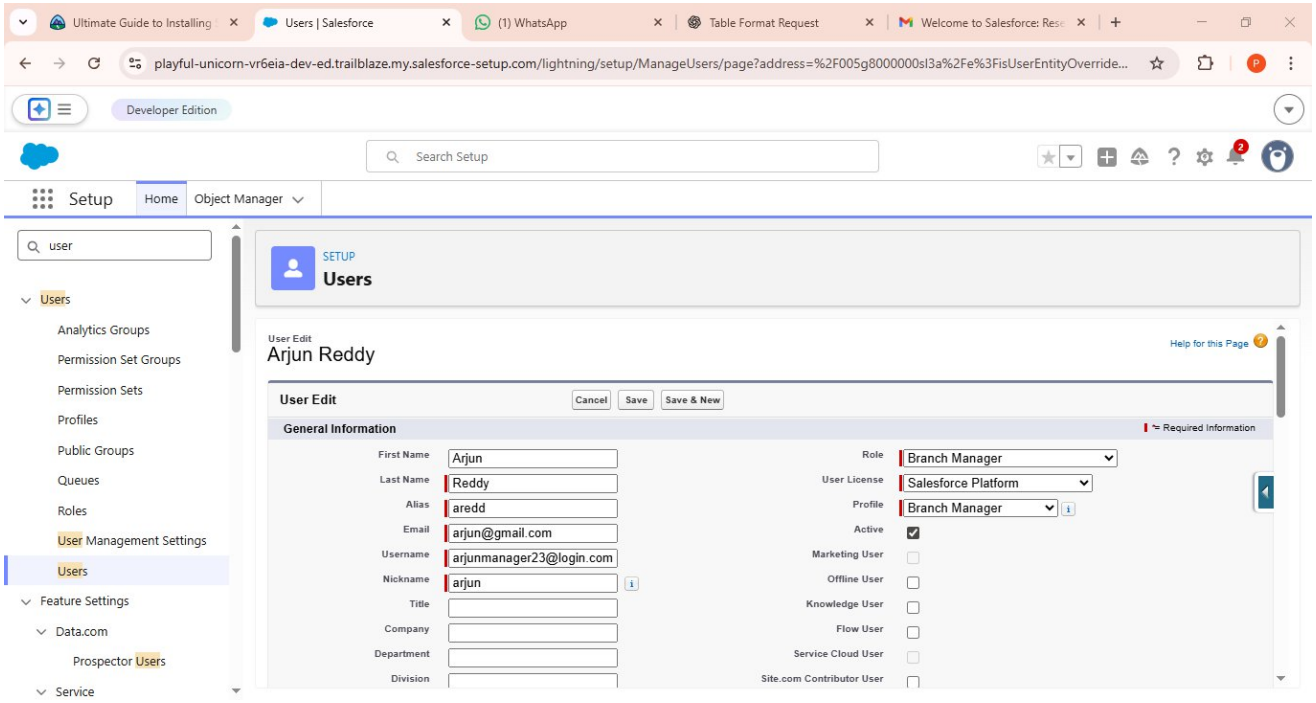
What is a user?

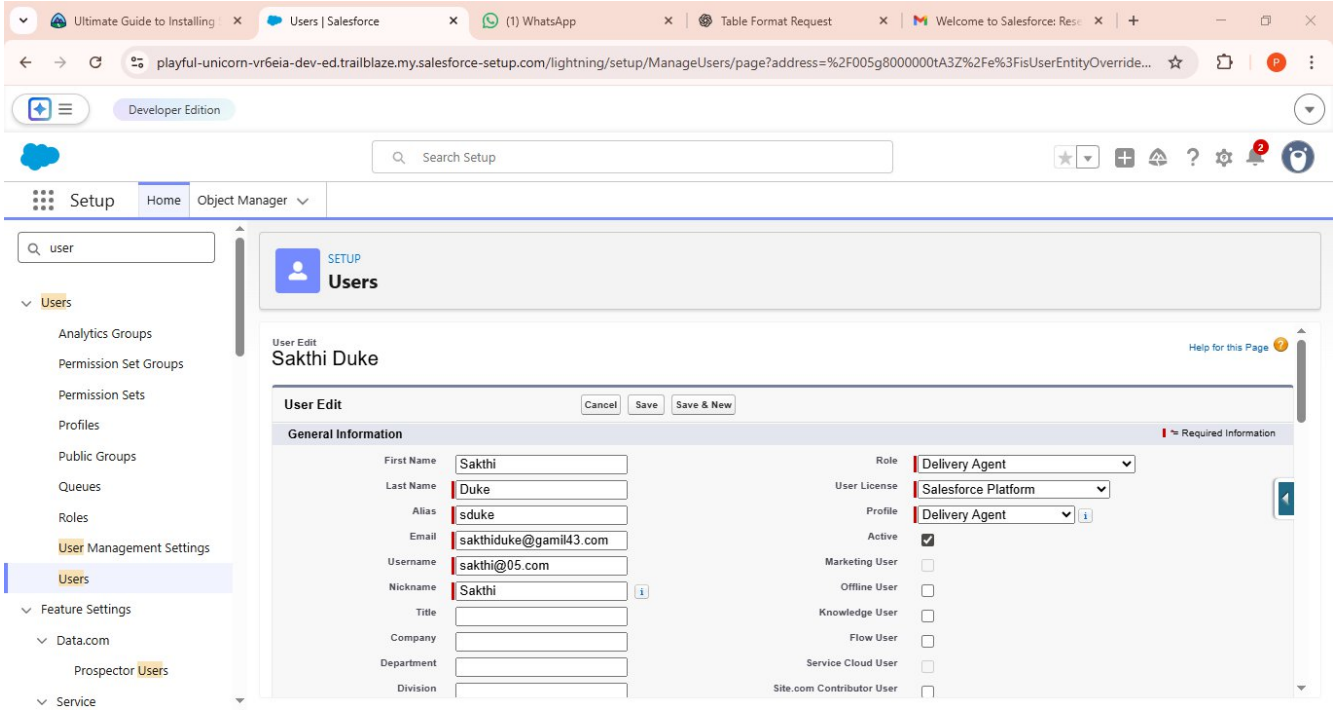
A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

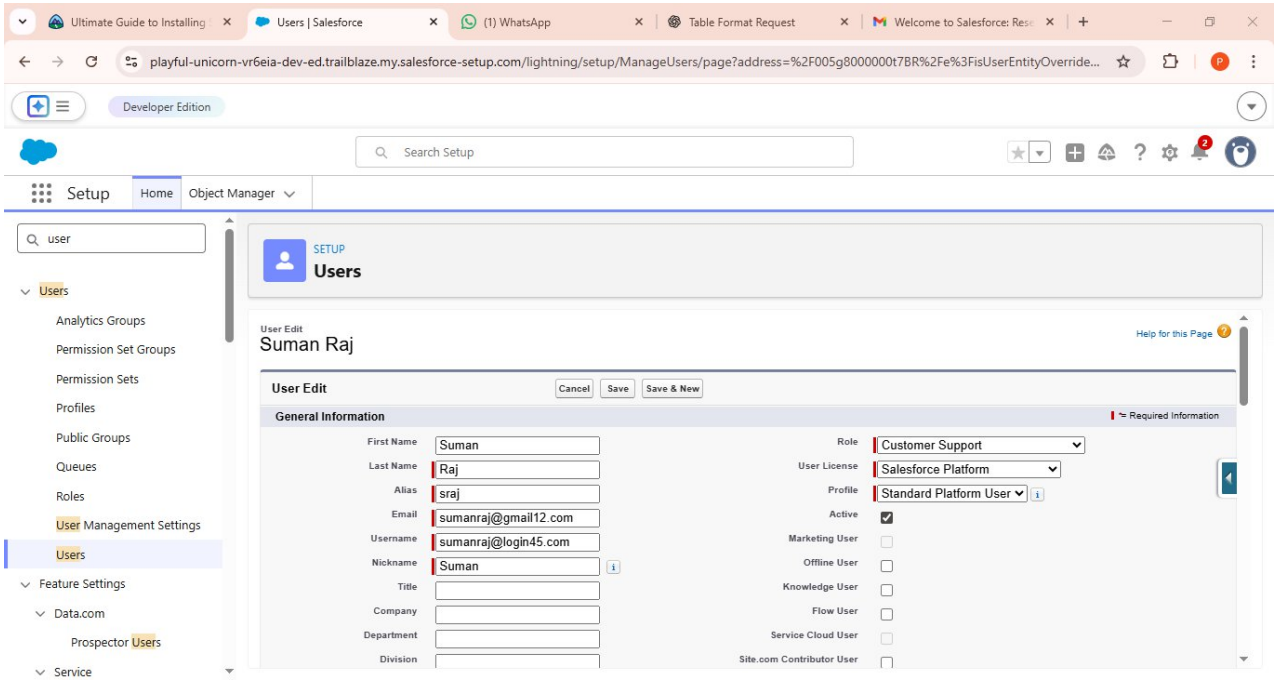
Activity 1: To Create a User



Activity 2: Creating Other Users







Milestone 15 : Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

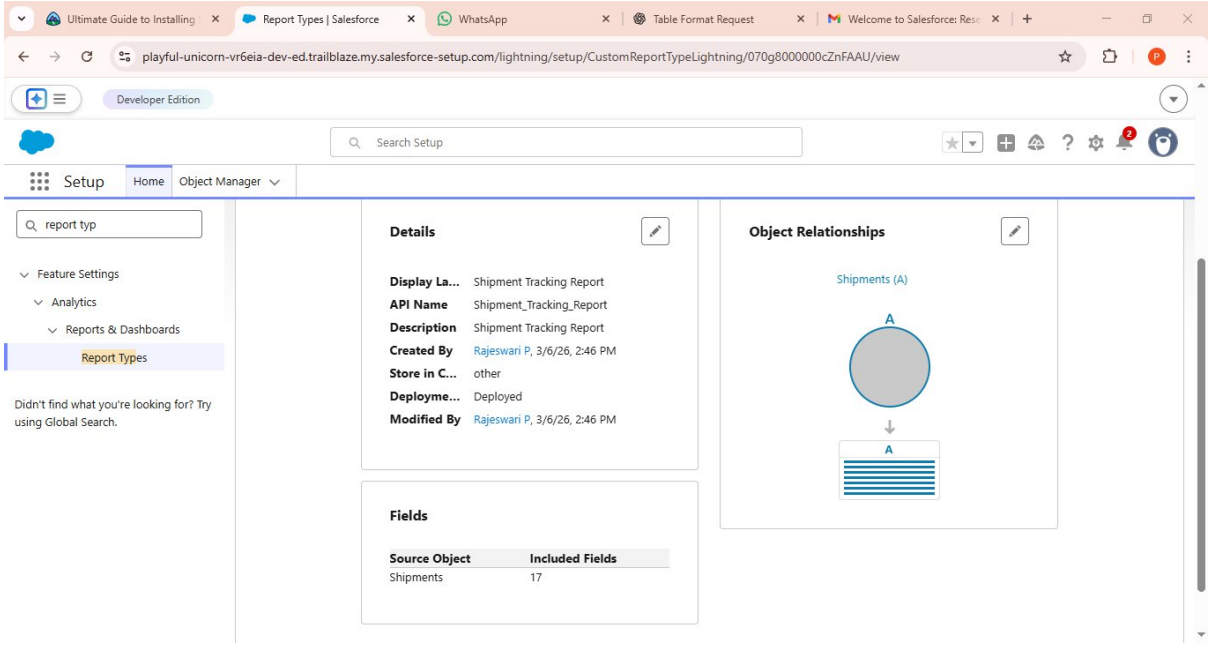
Step 1: Open Report Types

1. Click Gear Icon → Select Setup.

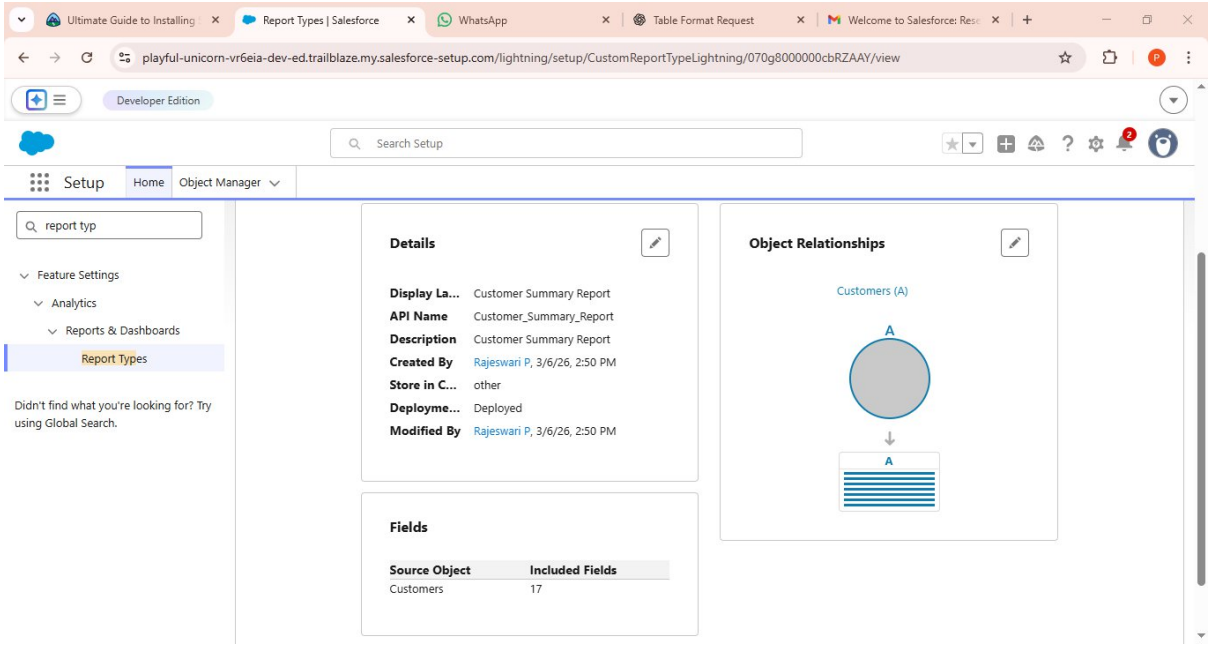
2. In Quick Find search Report Types.

3. Click New Custom Report Type.

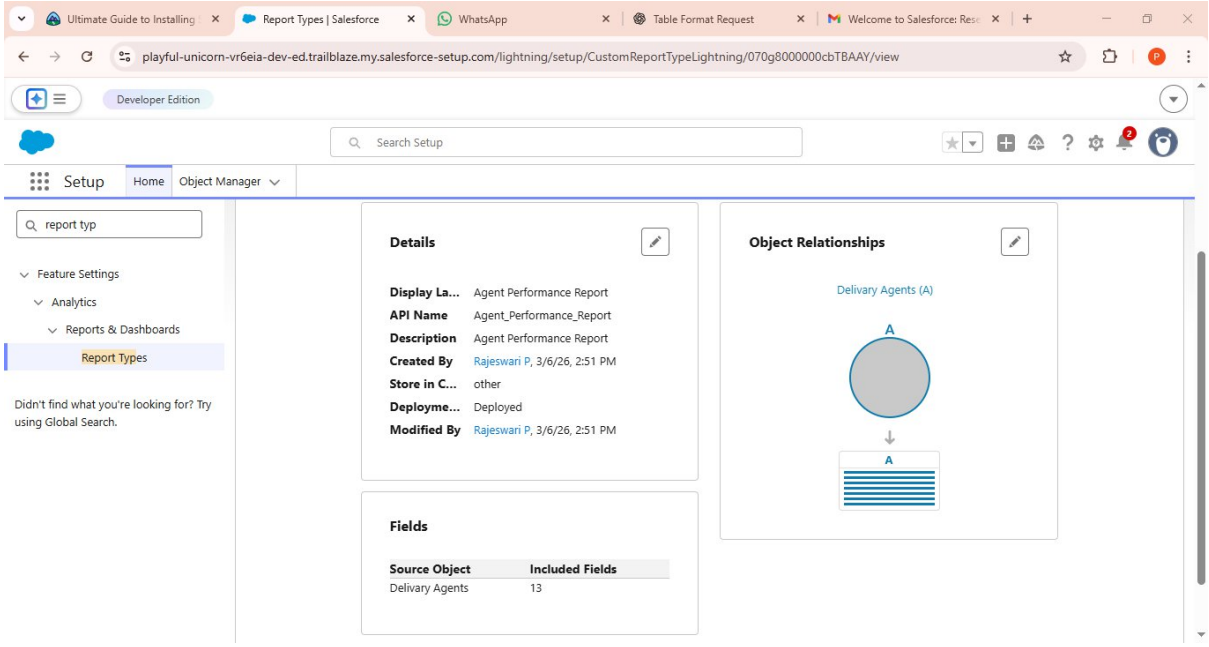
Step 2: Create Shipment Report Type



Step 3: Create Customer Report Type

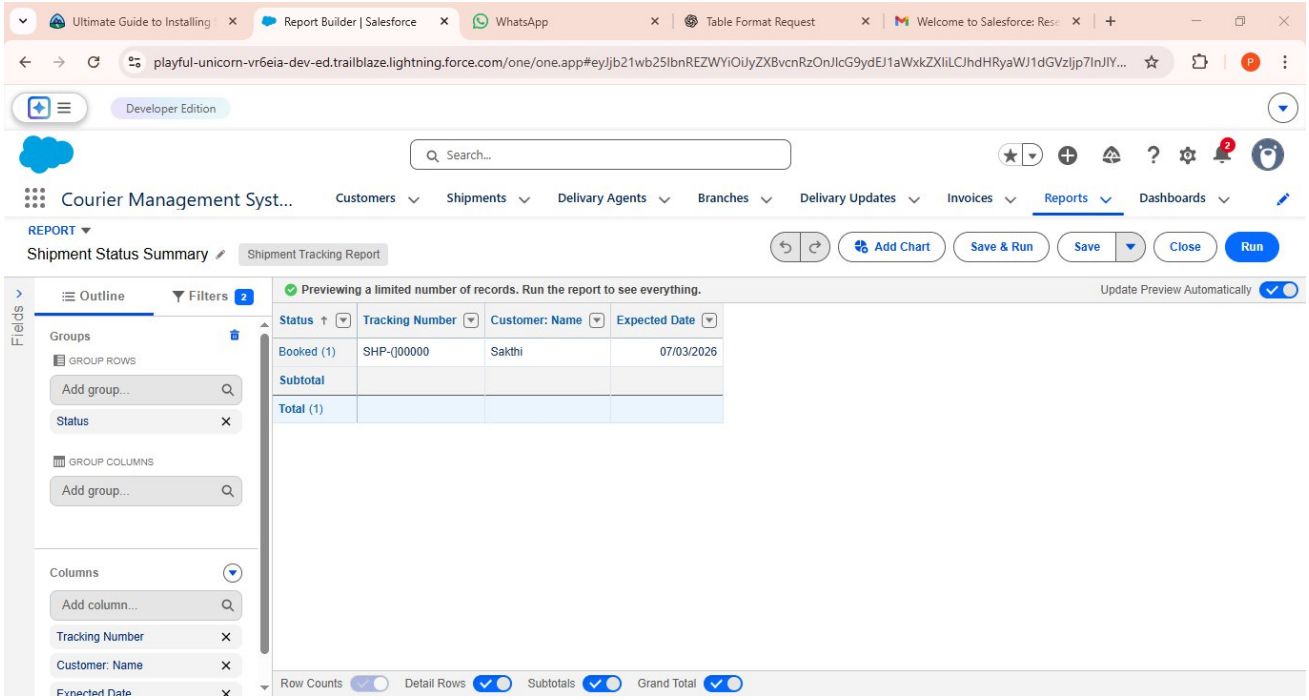


Step 4: Create Agent Performance Report Type

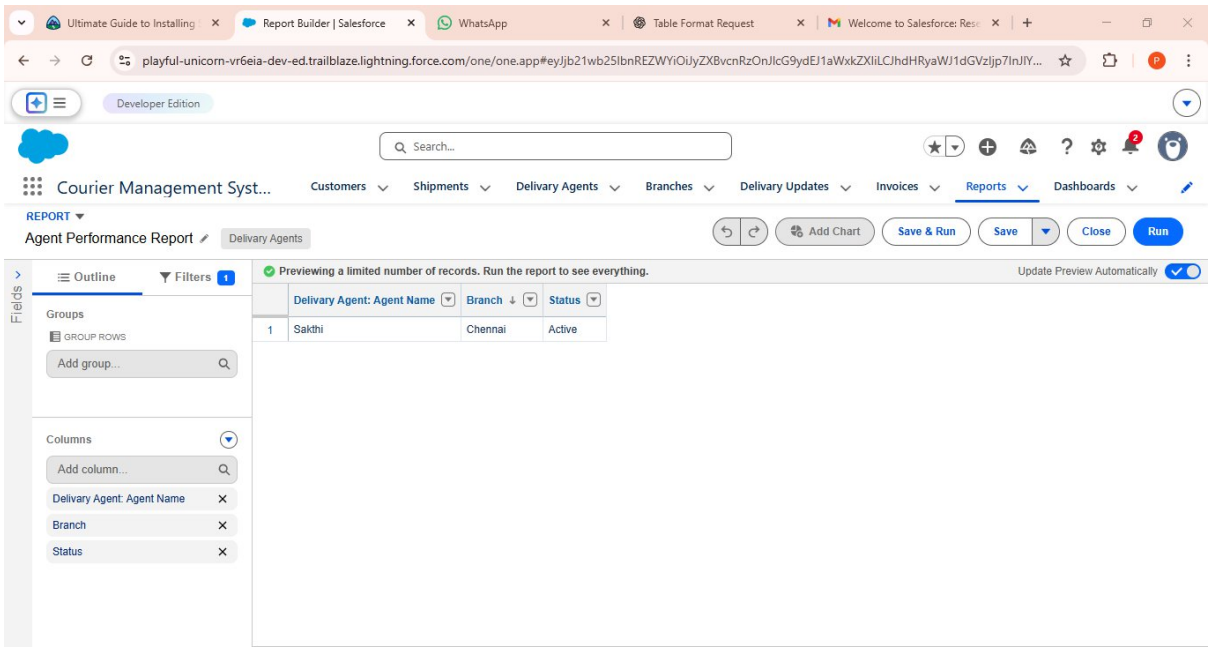


Activity 2: Create Reports

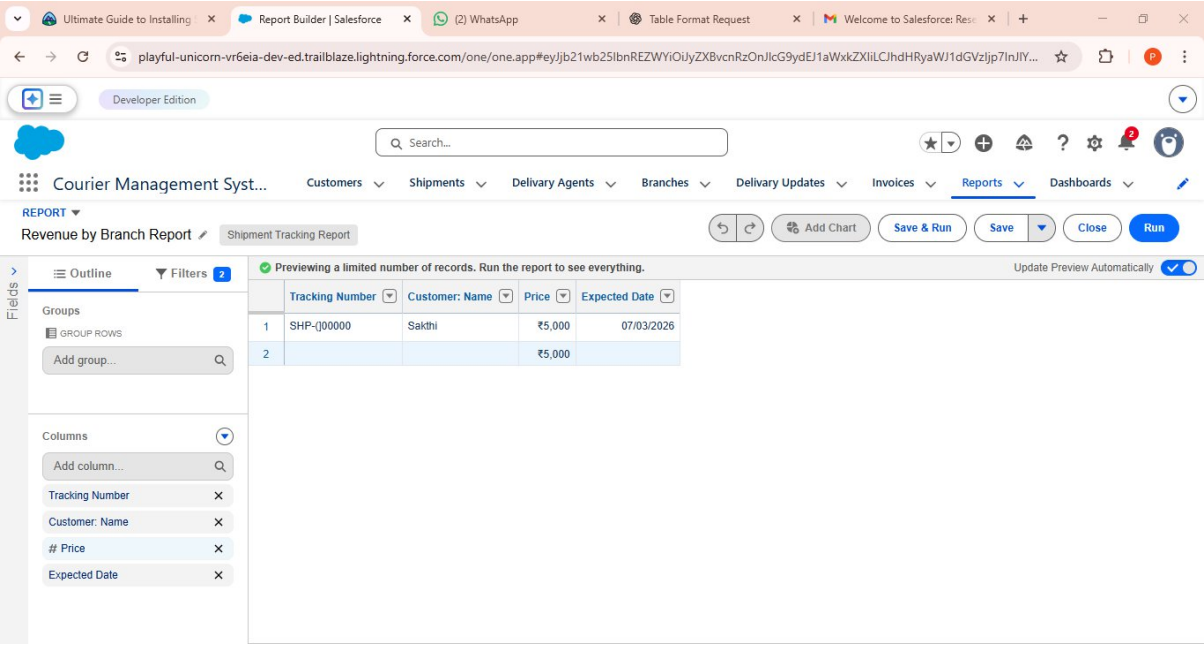
Report 1: Shipment Status Summary Report



Report 2: Agent Performance Report



Report 3: Revenue Report



Activity 3: Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create

Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

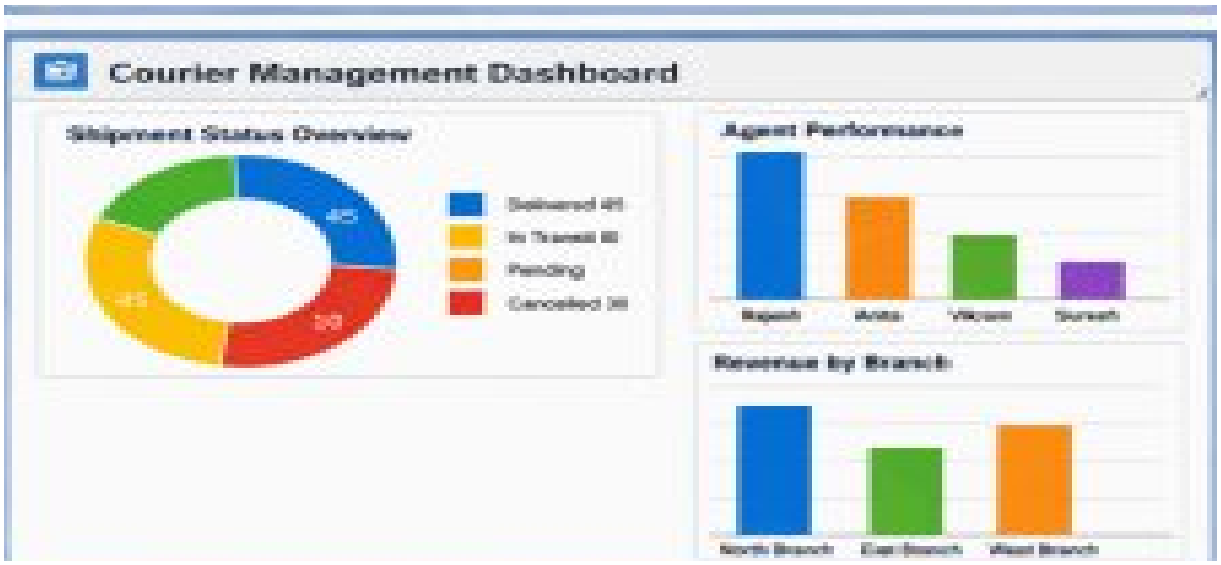
Component 2: Agent Performance (Bar Chart)

Component 3: Revenue by Branch (Column Chart)

Activity 5: Save & Activate Dashboard

Agent Performance Report
Total Shipments
165
Avg. Delivery Time
2.5 Days
60
45
35
25
Agent
Branch
Shipments
Avg. Delivery Time

Revenue by Branch Report
Jan
Feb
Mar
Apr
May
Branch
Revenue
Month


Courier Management Dashboard
Shipment Status Overview
Agent Performance
Revenue by Branch

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You